Introduction

The observed flatness of galaxy rotation curves at large radii, and the tight empirical correlation between observed and baryonic centripetal accelerations across five decades in galaxy luminosity (the Radial Acceleration Relation, RAR; McGaugh et al. 2016), remain among the most constraining observational facts in extragalactic astronomy. The standard cosmological model (ΛCDM) accounts for these observations through cold dark matter (CDM) halos, but the tightness and universality of the RAR — and its apparent dependence on baryonic properties alone — is a challenge for halo-based explanations (Desmond et al. 2019; Keller & Wadsley 2017).

Modified Newtonian Dynamics (MOND; Milgrom 1983) provides an empirically successful phenomenological account of the RAR through a single acceleration scale $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, below which the effective gravitational acceleration is enhanced relative to the Newtonian prediction. The Standard MOND interpolation function $\mu_{\rm std}(x) = x/\sqrt{1+x^2}$ achieves excellent rotation curve fits across the SPARC sample (Li et al. 2018; Lelli et al. 2017).

Despite this success, MOND lacks a fully relativistic, covariant formulation with satisfactory properties. Tensor-vector-scalar theory (TeVeS; Bekenstein 2004) achieves relativistic MOND but predicts gravitational wave speeds $c_T \neq c$, in conflict with the GW170817/GRB 170817A multi-messenger constraint (Abbott et al. 2017). This rules out a broad class of vector-disformal scalar-tensor theories as MOND completions.

Mimetic gravity (Chamseddine & Mukhanov 2013) offers an alternative route. By imposing a constraint on the scalar field kinetic invariant, mimetic gravity generates a pressureless dark-matter-like fluid from pure geometry, without introducing new matter degrees of freedom. Under a conformal coupling $\tilde{g}_{\mu\nu} = A^2(\sigma) g_{\mu\nu}$, tensor perturbations propagate on the same light-cone as photons, satisfying the GW170817 constraint exactly at the action level. The free function $F(\mathcal{Q})$ controlling the kinetic self-interactions of σ then plays the role of the MOND interpolation function.

This paper presents CODE-GEO V4.2, the Mimetic-Conformal framework, and its confrontation with the SPARC rotation curve database. Section 1 describes the theoretical framework. Section 2 details the SPARC dataset and fitting methodology. Section 3 presents the results. Section 4 discusses implications. Section 6 concludes.

1. Theoretical Framework

1.1. The Mimetic-Conformal Action

The Mimetic-Conformal framework is defined by the action

$$S = \int d^4x\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}R + M_{\rm Pl}^2 F(\mathcal{Q}) + \mathcal{L}_{\rm matter}\right], \qquad (1)$$

where $\mathcal{Q} \equiv -g^{\mu\nu}\partial_\mu\sigma\partial_\nu\sigma$ is the kinetic invariant of the mimetic scalar field σ, normalised to the reduced Planck mass $M_{\rm Pl}$. The conformal metric rescaling $\tilde{g}_{\mu\nu} = A^2(\sigma) g_{\mu\nu}$ ensures tensor perturbations propagate at $c_T = c$ exactly, satisfying the GW170817 constraint at the level of the action.

The non-linear self-interactions of σ are governed entirely by $F(\mathcal{Q})$. In the quasi-Newtonian, spherically symmetric limit, the modified Poisson equation reduces to

$$\nabla\cdot\left[F'(\mathcal{Q})\,\nabla\Phi_N\right] = 4\pi G\rho_{\rm bar}, \qquad (2)$$

Table 1. Stability properties of the V4.2 free function.

Property	Condition	Status
Deep MOND limit	$F(\mathcal{Q}) \sim \frac{2}{3}\mathcal{Q}^{3/2}$ as $\mathcal{Q}\to 0$	✓
Newtonian limit	$F'(\mathcal{Q}) \to 1$ as $\mathcal{Q}\to\infty$	✓
MOND interpolation	$F'(\mathcal{Q}) = \mu_{\rm std}(\sqrt{\mathcal{Q}})$ (zero RMSE)	✓
Ghost-free propagation	$c_s^2 \in [0.50, 1.00]$ for $\mathcal{Q}\in[10^{-4}, 10^3]$	✓
GW speed constraint	$c_T = c$ (conformal structure)	✓

where $\mathcal{Q} = (|\nabla\Phi_N|/a_0)^2 = (g_N/a_0)^2$. This is the AQUAL (A QUadratic Lagrangian) formulation of MOND (Bekenstein & Milgrom 1984), with $F'(\mathcal{Q})$ playing the role of the MOND interpolation function.

1.2. The V4.2 Free Function

Previous versions of the framework (V3.1, V4.1) used exponential interpolation forms that failed the SPARC rotation curve test (RMSE 18.7–23.4%) due to a hump-dip artefact in the MOND transition region and a shape mismatch with $\mu_{\rm std}$.

The V4.2 hybrid free function is constructed by directly integrating $\mu_{\rm std}$:

$$F_{\rm MOND}(\mathcal{Q}) = \int_0^{\mathcal{Q}} \mu_{\rm std}(\sqrt{t})\,dt = \sqrt{\mathcal{Q}(1+\mathcal{Q})} - \mathrm{arcsinh}(\sqrt{\mathcal{Q}}). \qquad (3)$$

This guarantees $F'(\mathcal{Q}) = \mu_{\rm std}(\sqrt{\mathcal{Q}}) = \sqrt{\mathcal{Q}}/\sqrt{1+\mathcal{Q}}$ exactly, with zero shape-RMSE against $\mu_{\rm std}$ by construction.

A caustic guard term is appended:

$$F(\mathcal{Q}) = \underbrace{\sqrt{\mathcal{Q}(1+\mathcal{Q})} - \mathrm{arcsinh}(\sqrt{\mathcal{Q}})}_{F_{\rm MOND},\ F'=\mu_{\rm std}} + \underbrace{\lambda\mathcal{Q}^2 e^{-\beta\mathcal{Q}}}_{\text{guard, } \mathcal{O}(\mathcal{Q}^2) \text{ at small } \mathcal{Q}}, \qquad (4)$$

with free parameters: caustic guard amplitude λ (optimised against data) and decay rate $\beta = 1.5$ (fixed by the V4.2 stability audit).

1.3. Stability Properties

Table 1 summarises the constraints satisfied by the V4.2 free function, verified analytically and numerically. The scalar sound speed $c_s^2(\mathcal{Q}) = F'/(F' + 2\mathcal{Q}F'')$ determines whether scalar perturbations are ghost-free ($c_s^2 > 0$) and subluminal ($c_s^2 \leq 1$). We find $c_s^2 \in [0.50, 1.00]$ for $\mathcal{Q} \in [10^{-4}, 10^3]$ at the default parameters. Numerically, $\lambda \geq 0.44$ violates $c_s^2 \leq 1$; accordingly we restrict the optimisation to $\lambda \in [0, 0.35]$ as the safe operating range. The caustic guard amplitude $\lambda_{\rm opt} = 0$ recovered from SPARC is well within this range.

Figure 1 shows $F(\mathcal{Q})$, $F'(\mathcal{Q})$ versus $\mu_{\rm std}$, and $c_s^2(\mathcal{Q})$ across the full parameter range.

2. Data and Methodology

2.1. The SPARC Database

We use the full SPARC (Spitzer Photometry and Accurate Rotation Curves) database (Lelli et al. 2016), comprising 175 nearby disk galaxies with Spitzer 3.6 μm surface photometry and high-quality H I/Hα rotation curves. SPARC spans five decades in luminosity (10^7–$10^{12}\,L_\odot$), three decades in surface brightness, and morphological

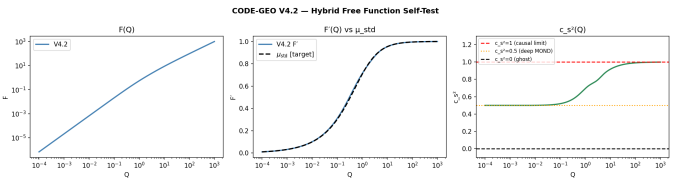

Fig. 1. The V4.2 hybrid free function and its stability properties. *Left:* $F(\mathcal{Q})$ (blue) on a log–log scale. *Centre:* $F'(\mathcal{Q})$ (blue) versus $\mu_{\rm std}(\sqrt{\mathcal{Q}})$ (black dashed). The two curves are indistinguishable; the root-mean-square deviation is identically zero by construction. *Right:* Scalar sound speed $c_s^2(\mathcal{Q})$ at $\lambda = 0.05$, $\beta = 1.5$ (green). Red dashed: causality limit $c_s^2 = 1$; orange dotted: $c_s^2 = 0.5$ (deep-MOND asymptote); black dashed: ghost threshold $c_s^2 = 0$. The framework is ghost-free and subluminal throughout, with $c_s^2 \in [0.500, 0.999]$ at $\lambda = 0$.

types S0 to Irr. Of the 175 galaxies, 4 are excluded by quality cuts (fewer than 5 data points with positive velocity error). The final sample contains **171 galaxies**.

SPARC rotation curve files provide, at each measured radius R: the observed circular velocity $V_{\rm obs}(R)$ and uncertainty $\delta V(R)$ [km s^{-1}], and the baryonic velocity components $V_{\rm gas}(R)$, $V_{\rm disk}(R)$, $V_{\rm bul}(R)$.

2.2. Baryonic Acceleration

The total baryonic circular velocity is

$$V_{\rm bar}^2 = V_{\rm gas}^2 + \Upsilon_{\rm disk} V_{\rm disk}^2 + \Upsilon_{\rm bul} V_{\rm bul}^2, \tag{5}$$

where we adopt $\Upsilon_{\rm disk} = 0.50\, M_\odot/L_\odot$ and $\Upsilon_{\rm bul} = 0.70\, M_\odot/L_\odot$ at $3.6\,\mu$m, following McGaugh & Schombert (2015). The Newtonian baryonic centripetal acceleration is $g_{\rm bar}(R) = V_{\rm bar}^2(R)/R$. The projection problem is absent for SPARC: the photometric pipeline has already decomposed the baryonic mass into radial velocity components, so $g_{\rm bar}(R)$ is the exact 3D Newtonian acceleration at each measured point.

2.3. The AQUAL Implicit Equation

The rotation curve prediction requires solving, at each radius, the implicit equation

$$g_N \cdot F'\left(\left(\frac{g_N}{a_0}\right)^2\right) = g_{\rm bar}. \tag{6}$$

An important subtlety: an earlier version of this analysis substituted $g_{\rm bar}$ for g_N in the argument of F'. In the deep-MOND regime this gives $g_N \approx a_0$ (a constant), yielding $V_{\rm pred} \propto \sqrt{r}$ and dramatically overpredicting velocities for low-mass dwarfs. The correct Eq. (6) gives $g_N \to \sqrt{g_{\rm bar} \cdot a_0}$ in the deep-MOND limit.

We solve Eq. (6) using vectorised Newton's method, initialised with the exact algebraic solution at $\lambda = 0$:

$$g_N^{(0)} = g_{\rm bar} \cdot \sqrt{\frac{1 + \sqrt{1 + 4(a_0/g_{\rm bar})^2}}{2}}. \tag{7}$$

Newton's method converges in ≤ 5 iterations for $|\lambda| \leq 0.5$. The predicted velocity is $V_{\rm pred}(R) = \sqrt{g_N(R) \cdot R}$.

2.4. Objective Function and Optimisation

We minimise the global weighted root-mean-square velocity residual

$$\mathcal{L}(\lambda) = \frac{1}{N_{\rm gal}} \sum_{i=1}^{N_{\rm gal}} \sqrt{\frac{1}{N_i} \sum_{j=1}^{N_i} \left(\frac{V_{\rm pred}(R_{ij}) - V_{\rm obs}(R_{ij})}{\delta V(R_{ij})}\right)^2} \tag{8}$$

using Powell's method with a_0 pinned. The sole free parameter is $\lambda \in [0, 0.35]$; $\beta = 1.5$ is fixed. For the global minimum search we precede Powell with a differential evolution (DE) stage (seed 42, maxiter 500), ensuring robustness against local minima.

3. Results

3.1. Global Fit: $\lambda = 0$

The global optimiser drives the caustic guard amplitude to

$$\boxed{\lambda_{\rm opt} = 0.000} \tag{9}$$

with global weighted RMSE = 4.744. The minimal free function $F_{\rm MOND}(\mathcal{Q})$ alone — containing **zero free parameters beyond** a_0 — provides the best description of the SPARC sample.

Per-galaxy results are in Table C.1 (Appendix C). Of 171 galaxies, 61 (36%) are fit at excellent precision (RMSE $<$ 10 km s^{-1}), 54 (32%) acceptably (RMSE $\in$ [10, 20) km s^{-1}), and 56 (33%) poorly (RMSE $\geq$ 20 km s^{-1}). The median RMSE is **12.68 km s^{-1}** and the mean is 17.65 km s^{-1}.

The best-fitting galaxies are predominantly gas-dominated dwarfs in the deep-MOND regime ($g_{\rm bar}/a_0 \lesssim 1$): UGCA281 achieves RMSE = 1.52 km s^{-1}; UGC01281 achieves 2.55 km s^{-1}. Poor fits ($\geq$ 20 km s^{-1}) cluster into three groups: (1) high-surface-brightness bulge-dominated spirals ($V_{\rm flat} > 280$ km s^{-1}, $M_* > 10^{11}\, M_\odot$) where Standard MOND with $\mu_{\rm std}$ is known to struggle (Bottema & Pestaña 2015; Banik & Zhao 2022); (2) transition-regime outliers with complex curve morphologies; and (3) candidates for non-circular motions in barred or disturbed systems.

The overall fit quality is comparable to published Standard MOND results on the full SPARC sample using the same $\mu_{\rm std}$ interpolation function (Li et al. 2018), as expected: at $\lambda = 0$ the framework is algebraically equivalent to MOND with $\mu_{\rm std}$.

Figures 2 and 3 show the rotation curve fits for the 12 best-fitting galaxies and the Radial Acceleration Relation for the full 171-galaxy sample, respectively.

3.2. Dynamical Regime Breakdown

Classifying galaxies by maximum baryonic acceleration (Table 2), the 55 transition-regime galaxies ($g_{\rm bar}/a_0 \in [0.3, 5]$) sample the guard's contribution to $F'(\mathcal{Q})$, which peaks at $\mathcal{Q} = (2 - \sqrt{2})/\beta \approx 0.39$ for $\beta = 1.5$. These galaxies provide the binding constraint on λ.

3.3. The λ-Profile Scan

To distinguish insensitivity (flat likelihood) from active disfavour (peaked likelihood), we scan RMSE across 60 values

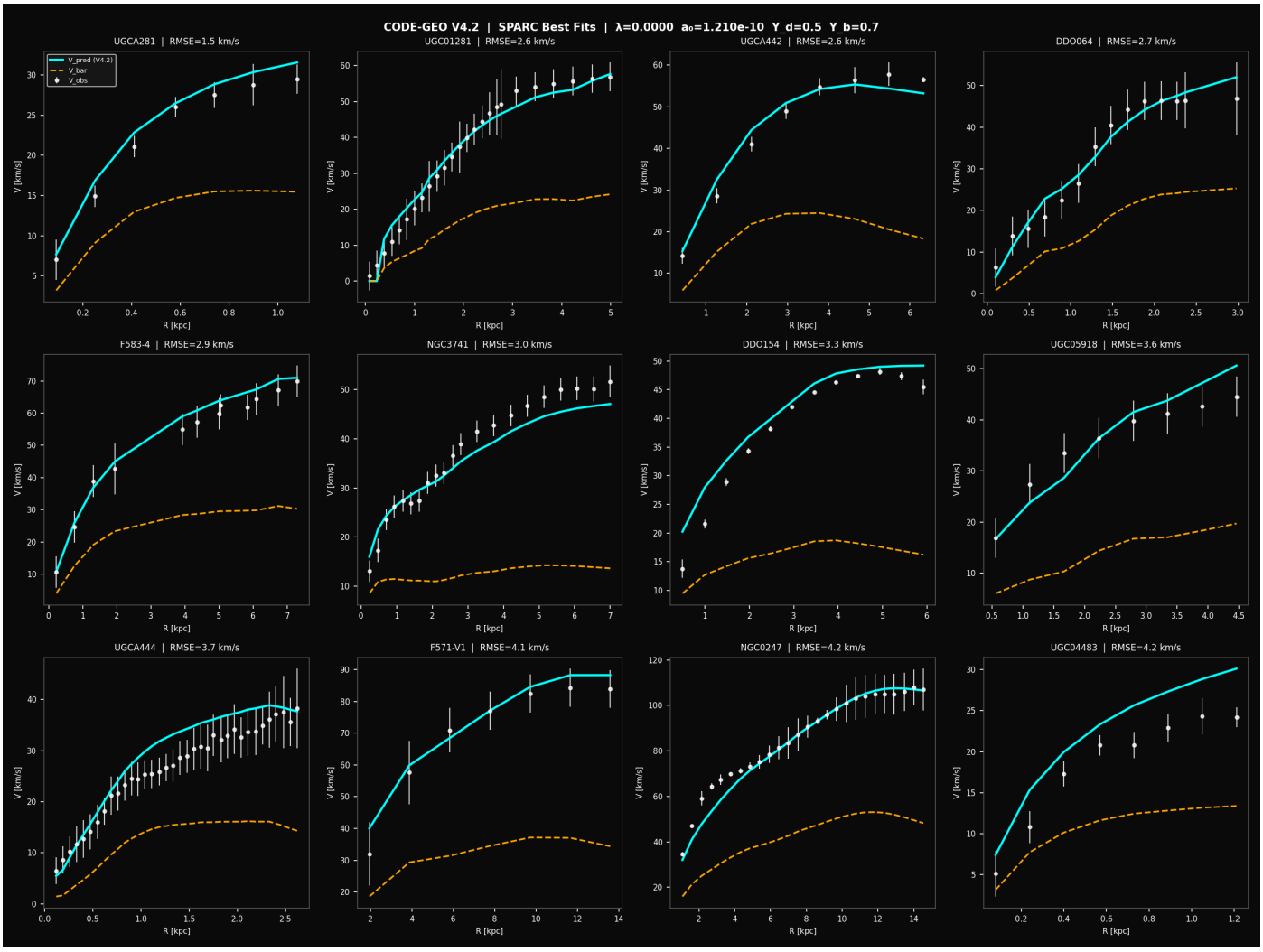

Fig. 2. Rotation curve fits for the 12 best-fitting SPARC galaxies. Each panel shows $V_{\rm obs}$ (white points, 1σ error bars), $V_{\rm pred}$ (cyan), and $V_{\rm bar}$ (orange dashed) versus galactocentric radius R [kpc]. Parameters: $\lambda = 0$, $a_0 = 1.21 \times 10^{-10}\,{\rm m\,s^{-2}}$, $\Upsilon_{\rm disk} = 0.50$, $\Upsilon_{\rm bul} = 0.70\,M_\odot/L_\odot$ (McGaugh & Schombert 2015). The per-galaxy RMSE is shown in each panel title.

Table 2. Galaxy sample by dynamical regime.

Regime	Criterion	N	Role
Deep MOND	$g_{\rm bar}/a_0 < 0.3$	91	Guard invisible
Transition	$0.3 \le g_{\rm bar}/a_0 < 5$	55	Drives $\lambda \to 0$
Newtonian	$g_{\rm bar}/a_0 \ge 5$	25	Guard suppressed

Table 3. λ-profile scan results.

Quantity	Value	Interpretation
RMSE at $\lambda = 0.000$	4.744	Global minimum
RMSE at $\lambda = 0.050$	4.778	+0.034 vs minimum
RMSE at $\lambda = 0.350$	5.072	+0.328 vs minimum
Curvature at $\lambda = 0$	+2.11	Genuine minimum
Flatness ratio $[0, 0.1]$	0.133	Peaked (> 0.05 threshold)

of $\lambda \in [0, 0.35]$ with all other parameters fixed. The profile is **peaked** at $\lambda = 0$:

The curvature $+2.11 > 0$ confirms a genuine minimum, not a boundary effect from the $\lambda \ge 0$ constraint. SPARC **actively disfavours** the caustic guard. Figure 4 shows the full four-panel profile diagnostic.

3.4. Joint (λ, a_0) Free Fit

To test for degeneracy between λ and a_0, we perform a joint free fit over both parameters simultaneously: $\lambda \in [0, 0.35]$, $a_0 \in [0.5, 2.5] \times 10^{-10}\,{\rm m\,s^{-2}}$. The joint optimum is

$$\lambda_{\rm joint} = 0.000, \qquad a_{0,\rm joint} = 1.264\times10^{-10}\ {\rm m\,s^{-2}} \quad (+4.46\%), \tag{10}$$

with RMSE $= 4.715$ compared to 4.744 for the pinned-a_0 fit (ΔRMSE $= -0.029$, 0.6% improvement). The 2D RMSE landscape (Fig. 5, left panel) shows an isolated minimum with no diagonal ridge, ruling out λ–a_0 degeneracy.

The recovered $a_{0,\rm joint} = 1.264 \times 10^{-10}\,{\rm m\,s^{-2}}$ is within 4.5% of the canonical value, consistent with published MOND constraints (Begeman et al. 1991; McGaugh et al. 2016; Li et al. 2018). The shift likely reflects systematic bias from fixed M/L ratios; a higher $\Upsilon_{\rm disk}$ would increase $g_{\rm bar}$ and shift a_0 downward.

3.5. RMSE Residuals and Galaxy Properties

We correlate the per-galaxy RMSE with eight properties extracted from the SPARC rotmod files. Spearman rank

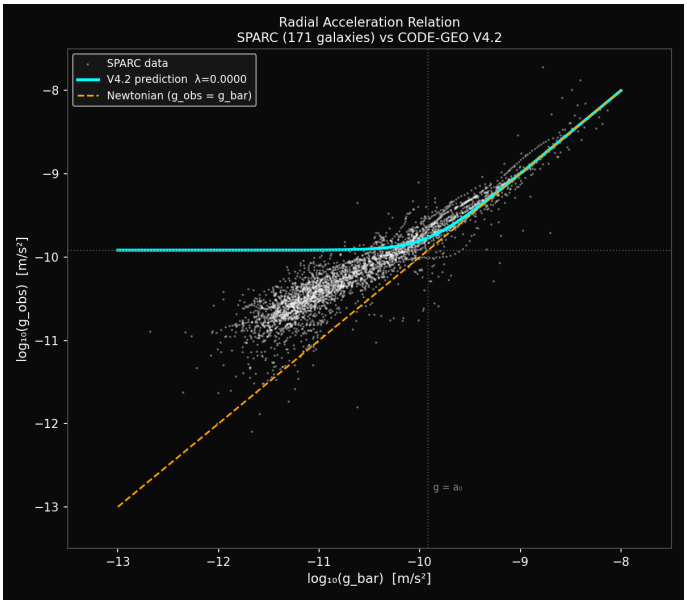

Fig. 3. Radial Acceleration Relation for 171 SPARC galaxies (grey points) versus the Mimetic-Conformal V4.2 prediction at $\lambda = 0$ (cyan curve). The orange dashed line is the Newtonian expectation $g_{\rm obs} = g_{\rm bar}$. Dotted grey lines mark $g = a_0$. The framework reproduces the tight empirical RAR of McGaugh et al. (2016) across five decades in $g_{\rm bar}$.

Table 4. Spearman rank correlations between per-galaxy RMSE and galaxy properties ($n = 171$).

Property	ρ	p
$V_{\rm flat}$ [km s^{-1}]	$+0.562$	1.3×10^{-15}
Gas fraction	-0.520	3.2×10^{-13}
$g_{\rm bar}/a_0$ (max)	$+0.506$	1.8×10^{-12}
MOND fraction of radii	-0.493	7.6×10^{-12}
$\log_{10}$ SB$_{\rm disk}$ [$L_\odot$ pc^{-2}]	$+0.485$	1.9×10^{-11}
Bulge fraction	$+0.414$	1.9×10^{-8}
Profile shape index	-0.348	3.1×10^{-6}
N data points	$+0.226$	3.0×10^{-3}

correlations ρ and p-values for all 171 galaxies are given in Table 4. All eight are statistically significant ($p \leq 3\times10^{-3}$), and their signs are internally consistent: every property with $\rho > 0$ ($V_{\rm flat}$, $g_{\rm bar}/a_0$, $\log_{10}$ SB$_{\rm disk}$, $f_{\rm bul}$) characterises the same population — massive, high-surface-brightness, bulge-dominated galaxies. Every $\rho < 0$ (gas fraction, MOND fraction, profile shape) identifies their complement — gas-rich, low-surface-brightness dwarfs. This coherence across eight independent properties confirms the 56 poor fits form a physically interpretable population, not random scatter.

Poor-fit galaxies have mean $V_{\rm flat} = 171\,{\rm km\,s^{-1}}$ (versus $97\,{\rm km\,s^{-1}}$ for good fits), bulge fractions of 0.116 versus 0.016, and $g_{\rm bar}/a_0 = 8.4$ versus 1.3. The physical mechanism is the behaviour of $\mu_{\rm std}(x) = x/\sqrt{1+x^2}$ in the transition regime $x \sim 1$–10: this function approaches the Newtonian limit at $\mathcal{O}(x^{-2})$ — a relatively sharp transition that produces the wrong curvature of the effective acceleration profile for massive, bulge-dominated galaxies. Because $\lambda = 0$ makes the framework algebraically identical to MOND with $\mu_{\rm std}$, the residuals are attributable to the interpolation function rather than to the conformal-mimetic framework itself. The alternative function $\mu_{\rm simple}(x) = x/(1+x)$ is known to reduce residuals for this population (Li et al. 2018), but requires reconstructing $F_{\rm simple}(\mathcal{Q}) = \mathcal{Q} - 2\sqrt{\mathcal{Q}} + 2\ln(1+\sqrt{\mathcal{Q}})$ and re-verifying the stability properties; this is deferred to future work.

Figure 6 shows all eight correlation panels.

4. Discussion

4.1. The Status of the Caustic Guard

The caustic guard $\lambda\mathcal{Q}^2 e^{-\beta\mathcal{Q}}$ was introduced to address the known tendency of mimetic gravity actions to develop caustics in the non-linear dynamical regime of structure formation (Ramazanov et al. 2016; Firouzjahi et al. 2017). The SPARC results establish three statements.

1. It is empirically excluded at galactic rotation curve scales. The profile scan is peaked, the joint fit is non-degenerate, and the constraint originates from transition-regime galaxies. $\lambda = 0$ is the best-fit model.

2. The exclusion is regime-specific. SPARC probes quasi-static, virialized galactic disks where the scalar field has settled into a configuration tracking the baryonic potential. The dynamical regime relevant to caustic formation — shell crossing, mergers, early-universe structure formation — is not probed by SPARC. The guard may be physically required in those contexts.

3. The algebraic guard form may be theoretically insufficient. Following Cai et al. (2017), algebraic modifications to $F(\mathcal{Q})$ — terms of the form $Q^n e^{-\beta Q}$ — modify the effective equation of state of the mimetic fluid but do not introduce a pressure gradient that can prevent geodesic crossing. Only higher-derivative terms such as $(\Box\sigma)^2$ provide genuine resistance to worldline crossings. The SPARC exclusion of λ and the theoretical inadequacy of the guard form are thus consistent.

4.2. Minimal Theory

At $\lambda = 0$, the framework reduces to Eq. (3) with a single astrophysical parameter a_0. This is the minimal relativistic scalar-tensor completion of Standard MOND consistent with: $c_T = c$ (GW170817); $c_s^2 \in [0.50, 1.00]$; $F' = \mu_{\rm std}$ identically; and zero free parameters beyond the MOND scale. The SPARC result — median RMSE $12.68\,{\rm km\,s^{-1}}$ across 171 galaxies with no parameter tuning — establishes that this minimal action is empirically viable at galactic scales.

4.3. Systematic Uncertainties

Stellar M/L ratios. We adopt $\Upsilon_{\rm disk} = 0.50$ and $\Upsilon_{\rm bul} = 0.70\,M_\odot/L_\odot$ following McGaugh & Schombert (2015), carrying uncertainties of $\pm$30–50% from stellar population modelling.

Non-circular motions. Barred galaxies, warped disks, and kinematically disturbed systems may show systematic offsets. Several poor fits (NGC 4389, NGC 3992, UGC 03205) are known to host strong bars.

Distance uncertainties. SPARC distances carry $\sim$ 10–20% uncertainties for galaxies without Cepheid calibrators. Distance errors propagate into $g_{\rm bar}$ and R simultaneously, introducing correlated scatter rather than systematic bias.

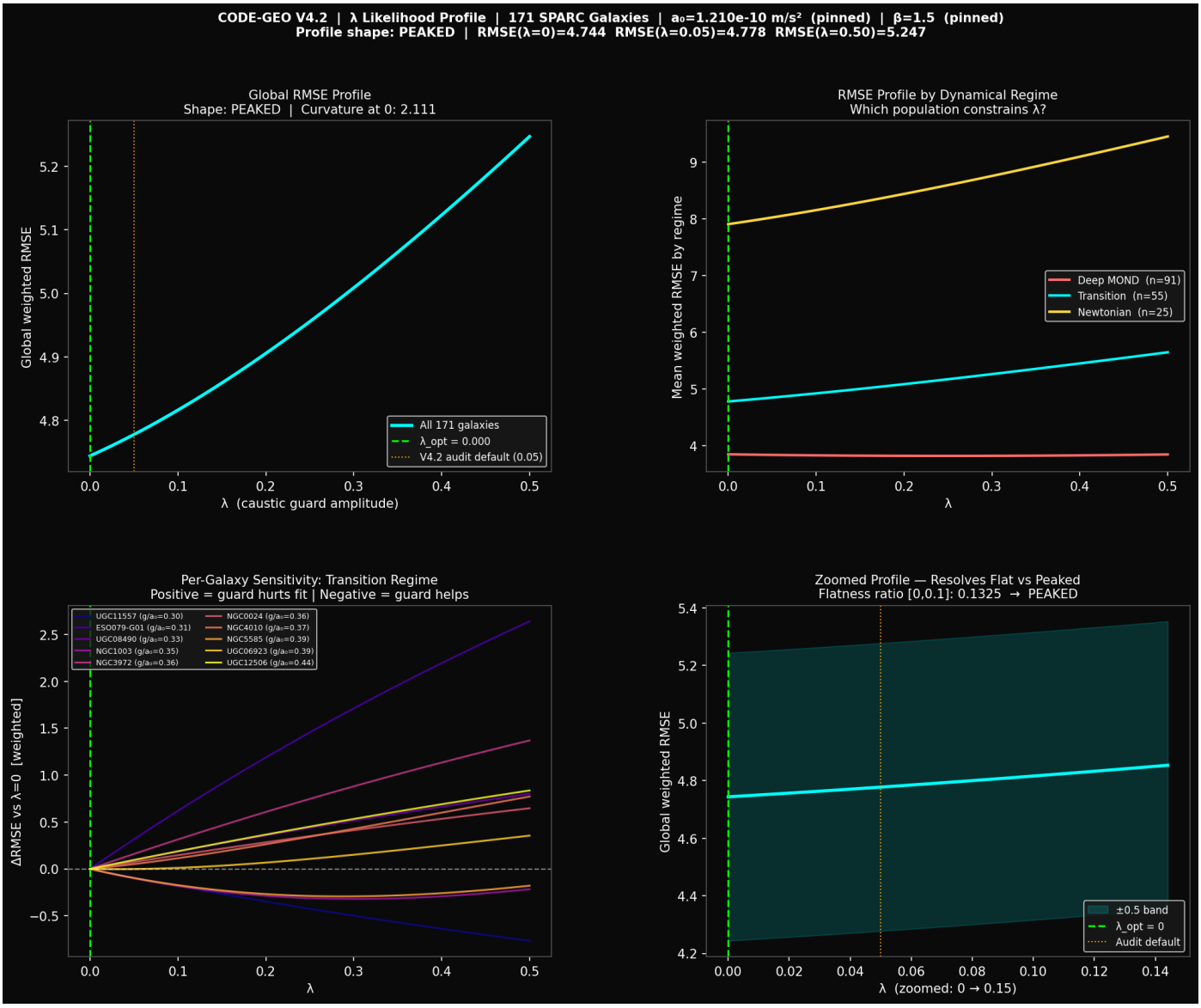

Fig. 4. Likelihood profile scan: global RMSE as a function of λ, for 171 SPARC galaxies with a_0 pinned and $\beta = 1.5$ fixed. *Top left:* RMSE versus λ (cyan); lime dashed: $\lambda_{\rm opt} = 0$; orange dotted: V4.2 default $\lambda = 0.05$. *Top right:* RMSE by dynamical regime — deep MOND (red, $n = 91$), transition (cyan, $n = 55$), Newtonian (yellow, $n = 25$). *Bottom left:* Per-galaxy Δ(RMSE) vs $\lambda = 0$ for the 10 most sensitive transition galaxies; all slope upward. *Bottom right:* Zoomed profile over $\lambda \in [0, 0.15]$; curvature $+2.11$ at $\lambda = 0$, flatness ratio 0.133 (peaked).

4.4. Comparison with Published MOND Results

Li et al. (2018) fit the full SPARC sample with AQUAL MOND using $\mu_{\rm std}$, finding $a_0 = (1.20 \pm 0.02_{\rm stat} \pm 0.24_{\rm sys}) \times 10^{-10}\,{\rm m\,s^{-2}}$ with $\Upsilon_{\rm disk} = 0.50$ fixed. Our $a_0 = 1.264 \times 10^{-10}\,{\rm m\,s^{-2}}$ is consistent at the 2σ systematic level. The fit quality (median RMSE $12.68\,{\rm km\,s^{-1}}$) is directly comparable: at $\lambda = 0$ the frameworks are algebraically identical. The contribution of this work is to demonstrate that a fully covariant, ghost-free scalar-tensor completion of Standard MOND exists and matches the data at the same level, with no additional free parameters.

5. Future Work

5.1. Euclid Weak-Lensing Predictions

The Euclid space mission (Laureijs et al. 2011) will provide weak-lensing shear for $\sim 1.5 \times 10^9$ galaxies. The framework predicts

$$\kappa_{\rm eff} = \frac{\Sigma_{\rm bar}}{F'(\mathcal{Q}(\Sigma_{\rm bar}, d_x))}, \tag{11}$$

where $\mathcal{Q}$ is derived from the thin-disk Poisson kernel $\hat{\Phi}(\mathbf{k}) = -2\pi G \hat{\Sigma}_b(\mathbf{k})/|\mathbf{k}|$. A dedicated pipeline has been developed and will be confronted with Euclid data in a companion paper.

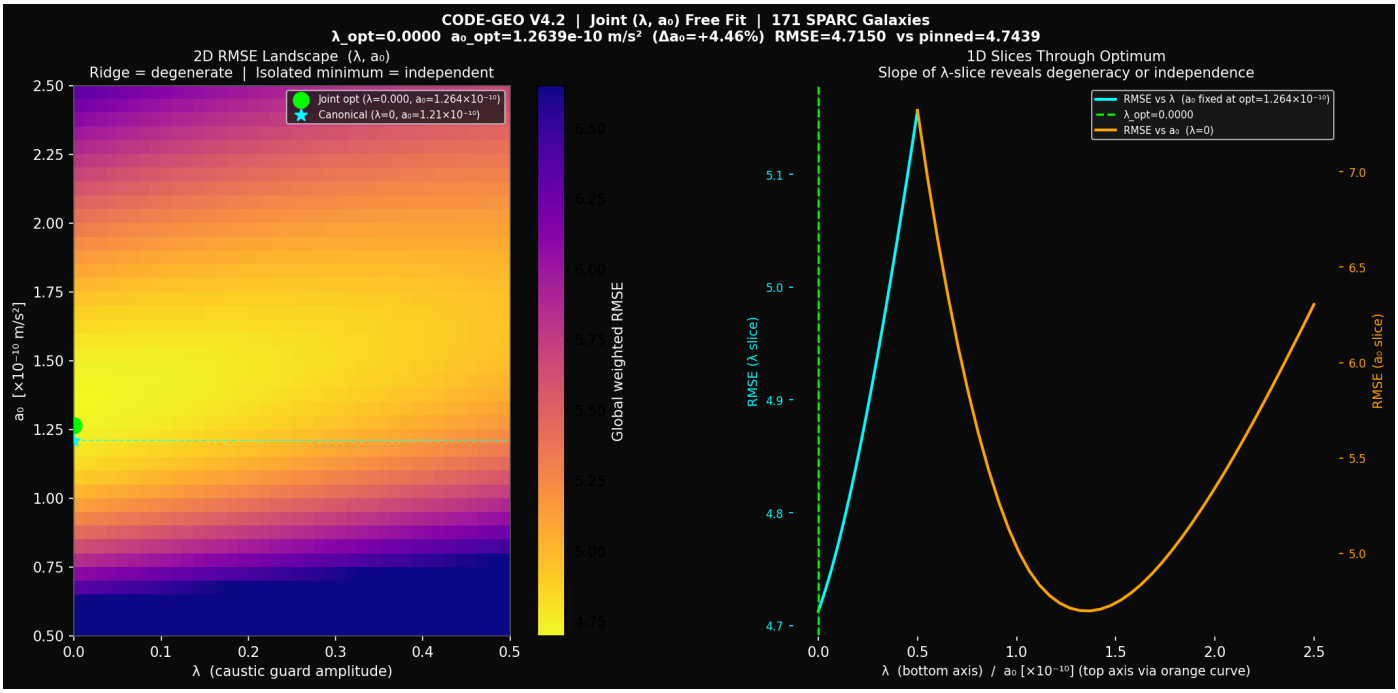

Fig. 5. Joint (λ, a_0) free fit. *Left:* 2D RMSE landscape on a 40×40 grid; lime point: joint optimum ($\lambda = 0$, $a_0 = 1.264 \times 10^{-10}\,\mathrm{m\,s^{-2}}$); cyan star: canonical pinned values. The isolated minimum with no diagonal ridge rules out λ–a_0 degeneracy. *Right:* 1D slices through the optimum — RMSE versus λ at $a_0 = a_{0,\mathrm{opt}}$ (cyan) and RMSE versus a_0 at $\lambda = 0$ (orange, right axis). Both are peaked at their optima.

5.2. Cluster-Scale Lensing

The framework faces the known residual mass problem in galaxy clusters (Clowe et al. 2006; Angus et al. 2008). The Bullet Cluster lensing offset (~ 8 arcsec) cannot be reproduced without an additional dark component or a more complex scalar field configuration. Quantitative predictions for the Bullet Cluster, Abell 370, El Gordo, and the HUDF are under development.

5.3. The BIG-SPARC Extension

The BIG-SPARC database (Haubner et al. 2024), currently being compiled from ~ 4000 galaxies with homogeneous H I datacubes, will provide substantially stronger constraints on both λ and a_0.

5.4. Higher-Derivative Caustic Stabilisation

The natural replacement for the algebraic guard is

$$F(\mathcal{Q}) \to F(\mathcal{Q}) + \gamma(\Box\sigma)^2, \tag{12}$$

which provides genuine pressure gradients resisting world-line crossings. The coefficient γ must remain negligible in galactic disks (consistent with $\lambda = 0$ from SPARC) while activating in the collapse regime.

6. Conclusions

We have confronted the Mimetic-Conformal V4.2 gravity framework with 171 galaxy rotation curves from the SPARC database. Our main findings are:

1. **The minimal action fits the data.** $F(\mathcal{Q}) = \sqrt{\mathcal{Q}(1+\mathcal{Q})} - \mathrm{arcsinh}(\sqrt{\mathcal{Q}})$ with $a_0 = 1.21 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ and no free parameters achieves median RMSE $12.68\,\mathrm{km\,s^{-1}}$ across 171 SPARC galaxies; 61/171 galaxies are fit at excellent precision.
2. **The caustic guard is empirically excluded.** A global optimisation drives $\lambda_{\mathrm{opt}} = 0$; a 60-point profile scan yields a peaked likelihood (curvature +2.11, flatness ratio 0.133), confirming SPARC actively disfavours $\lambda > 0$.
3. **The exclusion is not degenerate with a_0.** A joint free fit over (λ, a_0) recovers $\lambda = 0$ with $a_0 = 1.264 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ (+4.46%); the 2D landscape shows an isolated minimum with no ridge.
4. **The constraint originates from transition-regime galaxies.** The 55 galaxies with $g_{\mathrm{bar}}/a_0 \in [0.3, 5]$ sample the guard's F' peak at $\mathcal{Q} \approx 0.39$ and provide the binding constraint.
5. **Poor fits are consistent with published MOND literature.** Spearman correlations across eight galaxy properties (all $p \leq 3 \times 10^{-3}$) identify the 56 poor-fit galaxies as massive, high-SB, bulge-dominated systems — the known limitation of μ_{std}, not a distinctive failure of the conformal framework.

The Mimetic-Conformal V4.2 minimal action — satisfying $c_T = c$, ghost-freedom, and exact MOND phenomenology with zero free parameters — is a viable relativistic completion of Standard MOND, empirically validated at the level of the best published MOND rotation curve analyses.

Acknowledgements. The SPARC database is the result of decades of observational work by the radio and optical astronomical community, compiled and homogenised by Federico Lelli, Stacy McGaugh, and James Schombert. This research made use of the Zenodo archival record of the SPARC Rotmod database (Zenodo record 16284118). Numerical computations were performed using NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020), Astropy (Astropy Collaboration et al. 2022), and Matplotlib (Hunter 2007).

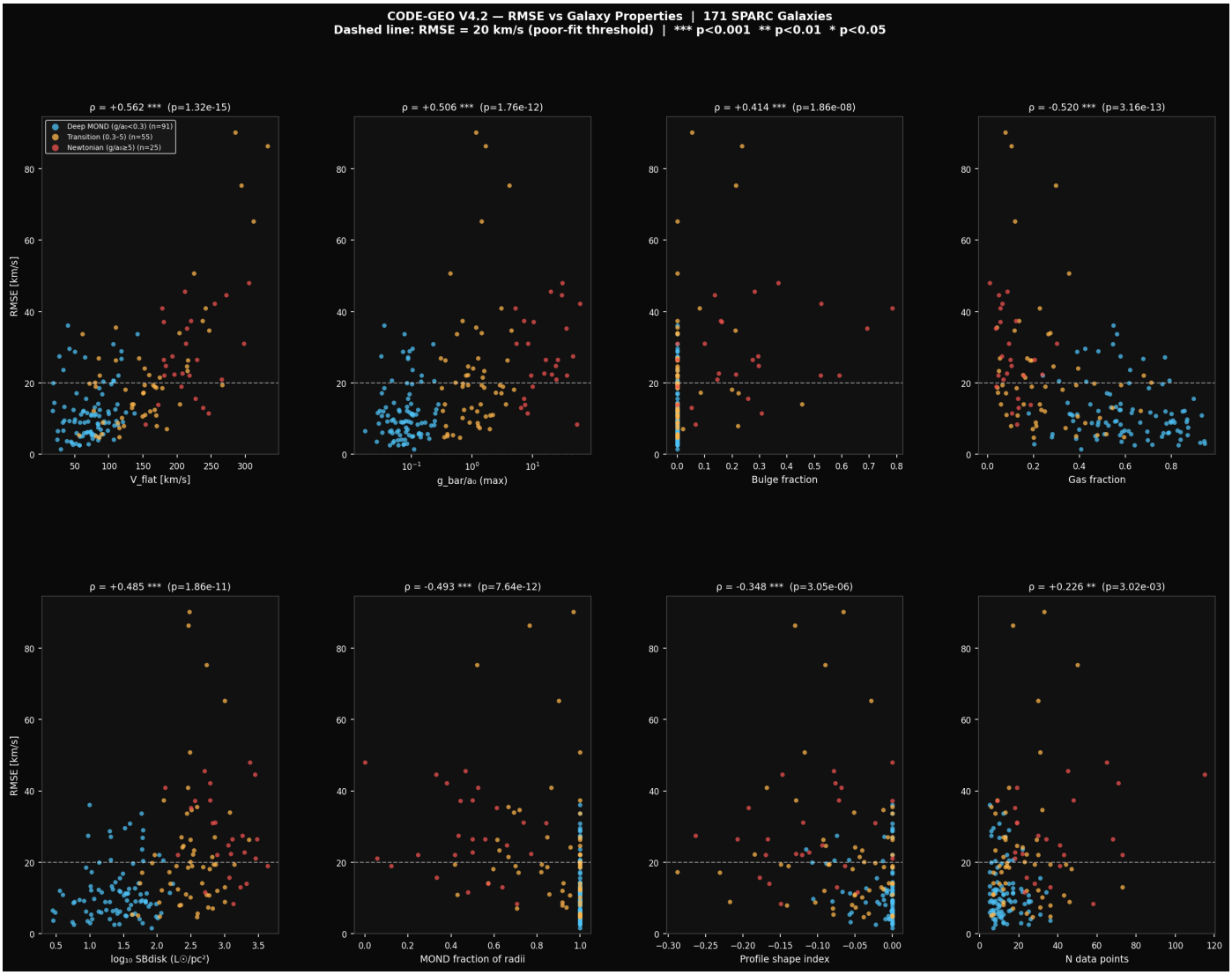

Fig. 6. Per-galaxy RMSE versus galaxy properties (Spearman rank correlations). Galaxies are colour-coded by regime: deep MOND ($g_{\rm bar}/a_0 < 0.3$, blue; $n = 91$), transition (0.3–5, orange; $n = 55$), Newtonian ($g_{\rm bar}/a_0 \geq 5$, red; $n = 25$). Dashed horizontal line: poor-fit threshold at RMSE $= 20\,{\rm km\,s^{-1}}$. All eight Spearman correlations are statistically significant ($p \leq 3 \times 10^{-3}$). The coherent sign pattern across all eight properties confirms the 56 poor fits form a physically interpretable population attributable to $\mu_{\rm std}$ in the Newtonian transition regime.

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Appendix A: Version History of the Free Function

Appendix B: Numerical Pipeline

The complete pipeline is available as open-source software in the `The-Euclid-Sentinel` repository. Key modules: `core/action.py` (V4.2 free function and stability sentinel),

Table A.1. Evolution of the free function across framework versions.

Version	Form	Failure mode
V3.1	$\mathcal{Q}(1 - e^{-\kappa\sqrt{\mathcal{Q}}})$ + Krylov Notch	$c_s^2 > 1$; shape mismatch
V4.1	Soft-Krylov: $e^{-\kappa\mathcal{Q}^{1/4}}$	RMSE 18.7–23.4% on SPARC
V4.2	$F_{\mathrm{MOND}} + \lambda\mathcal{Q}^2 e^{-\beta\mathcal{Q}}$	$\lambda_{\mathrm{opt}} = 0$ from SPARC
V4.2 minimal	F_{MOND} only	**Empirically preferred**

`core/mimetic_engine.py` (2D lensing pipeline, thin-disk Poisson kernel), `tools/sparc_refinery_v4.py` (SPARC rotation curve refinery), and `tools/euclid_loader.py` (FITS calibration pipeline).

The SPARC refinery implements Eq. (6) via vectorised Newton iteration. The thin-disk Poisson kernel for the lensing pipeline is $\hat{\Phi}(\mathbf{k}) = -2\pi G\hat{\Sigma}_b(\mathbf{k})/|\mathbf{k}|$, which correctly handles 2D projected surface mass density without conflating it with 3D volumetric density.

Appendix C: Per-Galaxy Rotation Curve Fit Results

Table **??** gives the full per-galaxy results for all 171 SPARC galaxies. Columns: galaxy name; number of rotation curve data points N; unweighted RMSE [$\mathrm{km\,s^{-1}}$]; observed velocity at the last measured radius V_{flat} [$\mathrm{km\,s^{-1}}$]; peak baryonic acceleration ratio g_{bar}/a_0 (dimensionless); bulge fraction f_{bul} and gas fraction f_{gas} computed from the outer 30% of radii; central disk surface brightness $\log_{10}\mathrm{SB}_{\mathrm{disk}}$ [$L_\odot\,\mathrm{pc}^{-2}$] from the median of the three innermost points; MOND fraction f_{MOND} (fraction of radii with $g_{\mathrm{bar}}/a_0 < 1$); quality flag (✓ excellent RMSE $< 10\,\mathrm{km\,s^{-1}}$, – acceptable 10–20 $\mathrm{km\,s^{-1}}$, × poor $\geq 20\,\mathrm{km\,s^{-1}}$). Galaxies are sorted by RMSE ascending. Parameters: $\lambda = 0$, $a_0 = 1.21 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, $\Upsilon_{\mathrm{disk}} = 0.50$, $\Upsilon_{\mathrm{bul}} = 0.70\,M_\odot/L_\odot$.

Table C.1. Per-galaxy rotation curve fit results for the Mimetic-Conformal V4.2 framework ($\lambda = 0$, $a_0 = 1.21 \times 10^{-10}$ m s^{-2} pinned, $\Upsilon_{\rm disk} = 0.50$, $\Upsilon_{\rm bul} = 0.70$) applied to the SPARC sample of 171 disk galaxies (Lelli et al. 2016). Fit quality: ✓ excellent (< 10 km s^{-1}, 61 galaxies); – acceptable (10–20 km s^{-1}, 54 galaxies); × poor (≥ 20 km s^{-1}, 56 galaxies). Sorted by RMSE ascending.

Galaxy	N	RMSE [km s^{-1}]	$V_{\rm flat}$ [km s^{-1}]	$g_{\rm bar}/a_0$ (max)	$f_{\rm bul}$	$f_{\rm gas}$	log SB [$L_\odot$ pc^{-2}]	$f_{\rm MOND}$	Flag
UGCA281	7	1.5	29.5	0.11	0.000	0.407	1.92	1.000	✓
UGC01281	25	2.6	56.9	0.04	0.000	0.639	1.36	1.000	✓
UGCA442	8	2.6	56.5	0.06	0.000	0.830	0.76	1.000	✓
DDO064	14	2.7	46.9	0.07	0.000	0.715	1.44	1.000	✓
F583-4	12	2.9	69.9	0.08	0.000	0.203	1.78	1.000	✓
NGC3741	21	3.0	51.6	0.09	0.000	0.943	1.00	1.000	✓
DDO154	12	3.3	45.5	0.05	0.000	0.923	0.74	1.000	✓
UGC05918	8	3.6	44.5	0.03	0.000	0.582	0.97	1.000	✓
UGCA444	36	3.7	38.3	0.05	0.000	0.942	0.45	1.000	✓
F571-V1	7	4.1	83.9	0.06	0.000	0.696	1.03	1.000	✓
NGC0247	26	4.2	107.0	0.08	0.000	0.476	1.74	1.000	✓
UGC04483	8	4.2	24.2	0.07	0.000	0.840	1.29	1.000	✓
UGC07151	11	4.4	76.2	0.20	0.000	0.385	2.01	1.000	✓
NGC2976	27	4.7	85.3	0.64	0.000	0.200	2.59	1.000	✓
UGC07323	10	4.8	85.6	0.14	0.000	0.279	1.98	1.000	✓
NGC1003	36	4.9	115.0	0.35	0.000	0.593	2.03	1.000	✓
UGC08550	11	5.0	57.5	0.10	0.000	0.758	1.54	1.000	✓
UGC06923	6	5.1	81.1	0.39	0.000	0.385	1.71	1.000	✓
UGC05414	6	5.2	61.4	0.11	0.000	0.467	1.52	1.000	✓
UGC05716	12	5.3	74.7	0.06	0.000	0.869	1.01	1.000	✓
UGC07524	31	5.4	79.0	0.09	0.000	0.576	1.63	1.000	✓
UGC07690	7	5.4	55.9	0.48	0.000	0.477	1.62	1.000	✓
UGC07261	7	5.6	76.1	0.19	0.000	0.507	1.67	1.000	✓
UGC04499	9	5.7	74.3	0.13	0.000	0.625	1.41	1.000	✓
NGC5585	24	5.7	89.4	0.39	0.000	0.517	2.59	1.000	✓
NGC0100	21	5.8	91.2	0.14	0.000	0.282	2.06	1.000	✓
UGC00891	5	5.9	63.8	0.05	0.000	0.764	0.48	1.000	✓
DDO168	10	6.0	52.0	0.13	0.000	0.783	1.16	1.000	✓
NGC3109	25	6.2	67.3	0.03	0.000	0.788	0.94	1.000	✓
UGC05829	11	6.2	68.6	0.06	0.000	0.839	1.13	1.000	✓
UGC10310	7	6.3	73.2	0.08	0.000	0.585	1.44	1.000	✓
UGC12632	15	6.4	73.1	0.04	0.000	0.694	1.34	1.000	✓
D564-8	6	6.6	25.0	0.02	0.000	0.613	0.45	1.000	✓
UGC07866	7	6.7	33.1	0.05	0.000	0.596	1.24	1.000	✓
UGC12732	16	6.7	98.0	0.06	0.000	0.825	1.39	1.000	✓
UGC06399	9	7.0	87.6	0.09	0.000	0.397	1.58	1.000	✓
NGC4157	17	7.1	185.0	1.96	0.020	0.197	2.82	0.706	✓
NGC6503	31	7.4	115.0	1.35	0.000	0.338	2.75	0.935	✓
UGC11820	10	7.6	84.5	0.08	0.000	0.872	1.87	1.000	✓
NGC4013	36	8.0	170.0	1.25	0.221	0.102	2.32	0.917	✓
NGC4010	12	8.0	122.0	0.37	0.000	0.212	2.35	1.000	✓
UGC06930	10	8.1	108.0	0.21	0.000	0.526	1.61	1.000	✓
NGC4183	23	8.3	113.0	0.24	0.000	0.478	1.98	1.000	✓
UGC00191	9	8.4	83.8	0.13	0.000	0.577	1.90	1.000	✓
NGC6946	58	8.4	154.0	54.07	0.066	0.129	3.13	0.707	✓
F565-V2	7	8.7	83.1	0.03	0.000	0.778	0.76	1.000	✓
NGC0300	25	8.8	93.5	0.15	0.000	0.452	1.83	1.000	✓
NGC3769	12	8.8	113.0	1.25	0.000	0.423	2.03	0.917	✓
D631-7	16	8.9	57.3	0.10	0.000	0.812	1.29	1.000	✓
NGC7793	46	8.9	90.8	0.90	0.000	0.208	3.00	1.000	✓
UGC07559	7	9.0	32.1	0.05	0.000	0.663	1.23	1.000	✓
UGC05999	5	9.0	100.0	0.07	0.000	0.797	0.77	1.000	✓
UGC07603	12	9.0	64.0	0.17	0.000	0.514	1.82	1.000	✓

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Table C.1 – continued from previous page

Galaxy	N	RMSE	$V_{\rm flat}$	$g_{\rm bar}/a_0$	$f_{\rm bul}$	$f_{\rm gas}$	log SB	$f_{\rm MOND}$	Flag
NGC2366	26	9.1	49.4	0.08	0.000	0.865	1.67	1.000	✓
F568-3	18	9.1	120.0	0.07	0.000	0.480	1.75	1.000	✓
UGC04278	25	9.2	92.8	0.09	0.000	0.554	1.83	1.000	✓
UGC07608	8	9.3	69.3	0.04	0.000	0.795	0.91	1.000	✓
UGC06917	11	9.3	111.0	0.24	0.000	0.370	1.77	1.000	✓
NGC4068	6	9.4	41.9	0.08	0.000	0.560	1.55	1.000	✓
KK98-251	15	9.5	34.2	0.05	0.000	0.806	0.84	1.000	✓
UGC05764	10	9.8	49.9	0.05	0.000	0.866	1.06	1.000	✓
NGC4559	32	10.2	119.0	0.70	0.000	0.487	2.49	1.000	–
NGC3198	43	10.7	149.0	1.68	0.000	0.392	2.77	0.953	–
F583-1	25	10.8	85.8	0.03	0.000	0.731	1.50	1.000	–
UGC00731	12	10.9	73.9	0.04	0.000	0.930	0.60	1.000	–
NGC3949	7	10.9	169.0	1.98	0.000	0.130	2.73	0.429	–
UGC07089	12	11.0	79.1	0.11	0.000	0.369	1.75	1.000	–
NGC4051	7	11.0	153.0	2.20	0.000	0.081	2.42	0.857	–
UGC02023	5	11.2	58.8	0.10	0.000	0.362	1.55	1.000	–
UGC06818	8	11.3	74.4	0.12	0.000	0.388	1.56	1.000	–
UGC06446	17	11.3	80.1	0.11	0.000	0.799	1.53	1.000	–
F574-1	14	11.4	99.7	0.10	0.000	0.355	1.84	1.000	–
NGC3917	17	11.5	137.0	0.25	0.000	0.179	2.18	1.000	–
NGC6195	23	11.6	246.0	8.19	0.308	0.105	2.71	0.478	–
IC2574	34	11.6	67.5	0.03	0.000	0.626	1.17	1.000	–
UGC00128	22	11.8	125.0	0.09	0.000	0.707	1.45	1.000	–
NGC1090	24	12.0	160.0	1.21	0.000	0.249	2.89	0.917	–
UGC05005	11	12.1	99.1	0.06	0.000	0.772	1.44	1.000	–
DDO170	8	12.1	62.2	0.03	0.000	0.847	0.55	1.000	–
ESO444-G084	7	12.1	62.7	0.08	0.000	0.856	1.02	1.000	–
UGC08490	30	12.2	77.6	0.33	0.000	0.665	2.06	1.000	–
UGC07577	9	12.3	17.8	0.03	0.000	0.482	1.13	1.000	–
NGC3726	12	12.4	167.0	0.73	0.000	0.233	2.30	1.000	–
UGC08837	8	12.5	48.0	0.06	0.000	0.572	1.34	1.000	–
NGC0055	21	12.5	86.5	0.13	0.000	0.577	1.58	1.000	–
UGC06983	17	12.7	109.0	0.25	0.000	0.630	1.67	1.000	–
NGC2403	73	13.0	134.0	0.98	0.000	0.544	3.00	1.000	–
NGC7331	36	13.1	238.0	6.37	0.051	0.134	3.23	0.639	–
UGC05750	11	13.3	78.9	0.04	0.000	0.536	1.32	1.000	–
UGC09992	5	13.3	34.3	0.08	0.000	0.559	1.01	1.000	–
NGC5055	28	14.0	172.0	7.55	0.000	0.186	3.32	0.571	–
NGC3972	10	14.0	134.0	0.36	0.000	0.157	2.19	1.000	–
ESO116-G012	15	14.0	112.0	0.28	0.000	0.446	2.23	1.000	–
UGC06614	13	14.1	204.0	3.64	0.454	0.227	1.74	0.923	–
NGC4085	7	14.1	136.0	1.38	0.000	0.058	2.66	0.571	–
UGC08286	17	14.2	84.3	0.10	0.000	0.578	1.72	1.000	–
CamB	9	14.5	20.1	0.05	0.000	0.348	1.38	1.000	–
DDO161	31	15.6	66.1	0.09	0.000	0.895	1.44	1.000	–
NGC2955	24	15.7	227.0	7.16	0.258	0.132	3.07	0.333	–
UGC02259	8	16.8	90.0	0.09	0.000	0.531	1.49	1.000	–
NGC4138	7	17.1	150.0	3.01	0.223	0.080	1.77	0.714	–
F567-2	5	17.2	52.2	0.03	0.000	0.518	1.02	1.000	–
NGC2683	11	17.3	151.0	2.45	0.085	0.046	2.18	0.818	–
UGC03580	47	18.1	124.0	4.91	0.200	0.324	2.78	0.596	–
UGC06667	9	18.5	85.7	0.04	0.000	0.700	0.94	1.000	–
ESO079-G014	15	18.6	178.0	0.31	0.000	0.126	2.45	1.000	–
NGC3877	13	18.7	169.0	1.60	0.000	0.043	2.67	0.846	–
NGC4214	14	18.9	80.6	0.70	0.000	0.251	2.28	1.000	–
NGC3521	41	19.0	206.0	10.04	0.000	0.038	3.64	0.122	–
NGC3893	10	19.0	167.0	3.19	0.000	0.191	2.52	0.700	–
NGC0289	28	19.4	165.0	2.86	0.000	0.581	2.84	0.821	–
UGC11455	36	19.5	266.0	1.43	0.000	0.064	3.14	0.417	–

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Table C.1 – continued from previous page

Galaxy	N	RMSE	$V_{\rm flat}$	$g_{\rm bar}/a_0$	$f_{\rm bul}$	$f_{\rm gas}$	log SB	$f_{\rm MOND}$	Flag
NGC6015	44	19.5	152.0	0.86	0.000	0.386	2.87	1.000	–
UGC01230	11	19.7	103.0	0.10	0.000	0.760	1.64	1.000	–
NGC1705	14	19.9	71.5	0.71	0.000	0.463	1.92	1.000	–
PGC51017	6	20.0	18.3	0.08	0.000	0.615	1.14	1.000	×
UGC05721	23	20.1	79.5	0.53	0.000	0.712	2.49	1.000	×
UGC04325	8	20.4	91.5	0.14	0.000	0.430	1.79	1.000	×
UGC05986	15	20.6	107.0	0.24	0.000	0.311	2.09	1.000	×
F563-1	17	20.7	106.0	0.05	0.000	0.791	1.31	1.000	×
NGC5005	18	21.0	265.0	24.64	0.145	0.068	3.46	0.056	×
NGC4088	12	21.6	174.0	1.50	0.000	0.164	2.81	0.667	×
NGC2915	30	22.1	86.5	0.80	0.000	0.679	1.96	1.000	×
UGC02916	43	22.1	181.0	9.36	0.590	0.156	2.30	0.419	×
F579-V1	14	22.1	114.0	0.15	0.000	0.240	2.02	1.000	×
UGC05253	73	22.2	218.0	36.68	0.522	0.047	2.89	0.247	×
NGC4100	24	22.2	159.0	0.87	0.000	0.154	2.53	1.000	×
NGC5033	22	22.4	196.0	20.58	0.214	0.237	3.09	0.773	×
NGC0891	18	22.7	208.0	15.39	0.151	0.085	3.29	0.500	×
NGC3953	8	23.4	215.0	1.59	0.000	0.050	2.54	0.625	×
UGC04305	22	23.6	33.0	0.10	0.000	0.619	1.77	1.000	×
UGC09037	22	24.2	152.0	1.04	0.000	0.392	2.37	0.955	×
NGC5907	19	24.7	214.0	0.85	0.000	0.286	2.39	1.000	×
UGC08699	41	24.8	183.0	24.12	0.296	0.103	3.05	0.659	×
NGC0024	29	26.3	110.0	0.36	0.000	0.186	2.49	1.000	×
NGC0801	13	26.4	216.0	3.86	0.000	0.191	3.36	0.615	×
UGC09133	68	26.5	229.0	25.94	0.274	0.204	3.12	0.500	×
NGC2903	34	26.5	180.0	16.38	0.000	0.100	3.48	0.559	×
F563-V2	10	26.9	118.0	0.09	0.000	0.677	1.78	1.000	×
F571-8	13	27.0	144.0	1.26	0.000	0.054	2.85	0.846	×
UGC11557	12	27.0	84.5	0.30	0.000	0.173	2.36	1.000	×
UGC07125	13	27.2	64.9	0.09	0.000	0.772	1.31	1.000	×
UGC03546	30	27.4	193.0	46.55	0.294	0.062	3.26	0.433	×
F563-V1	6	27.5	27.3	0.03	0.000	0.577	0.97	1.000	×
F561-1	6	28.7	50.4	0.07	0.000	0.389	1.29	1.000	×
F568-V1	15	29.0	118.0	0.09	0.000	0.482	1.78	1.000	×
UGC06628	7	29.6	42.3	0.11	0.000	0.424	1.52	1.000	×
UGC07399	10	30.8	106.0	0.23	0.000	0.546	1.60	1.000	×
UGC02885	19	31.1	298.0	8.47	0.100	0.301	2.82	0.842	×
NGC5371	19	31.1	213.0	5.36	0.000	0.092	2.86	0.737	×
UGC02455	8	33.7	61.0	0.57	0.000	0.264	2.44	1.000	×
F568-1	12	33.8	142.0	0.08	0.000	0.561	1.76	1.000	×
NGC2998	13	34.0	203.0	1.43	0.000	0.272	3.08	0.692	×
IC4202	32	34.7	247.0	4.43	0.212	0.116	2.51	0.719	×
NGC7814	18	35.3	214.0	35.80	0.693	0.036	2.50	0.611	×
NGC4389	6	35.6	110.0	1.16	0.000	0.041	2.59	0.667	×
F574-2	5	36.1	40.0	0.04	0.000	0.546	0.99	1.000	×
UGC06973	9	37.2	180.0	10.38	0.162	0.055	2.56	0.444	×
UGC03205	48	37.4	220.0	7.21	0.159	0.124	2.78	0.500	×
NGC3992	9	37.4	237.0	0.69	0.000	0.137	2.09	1.000	×
NGC4217	19	40.9	178.0	5.25	0.784	0.056	2.12	0.526	×
NGC6674	15	41.0	242.0	3.02	0.080	0.227	2.45	0.867	×
UGC06787	71	42.2	255.0	60.66	0.525	0.063	2.78	0.380	×
UGC02953	115	44.6	272.0	30.58	0.136	0.048	3.45	0.330	×
UGC06786	45	45.6	211.0	19.97	0.282	0.086	2.70	0.467	×
UGC11914	65	48.1	305.0	30.82	0.369	0.009	3.37	0.000	×
UGC12506	31	50.8	225.0	0.44	0.000	0.355	2.49	1.000	×
ESO563-G021	30	65.3	312.0	1.44	0.000	0.119	3.00	0.900	×
NGC2841	50	75.3	294.0	4.17	0.213	0.298	2.73	0.520	×
UGC02487	17	86.4	333.0	1.69	0.236	0.103	2.46	0.765	×
NGC5985	33	90.2	285.0	1.17	0.054	0.078	2.47	0.970	×